

DASC 5304 – Machine Learning Midway Progress Report

Project Title: Robustness Enhancement in Image Classification via Data Augmentation

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Project Overview

In this project, we study how different data augmentation methods affect the accuracy and robustness of a deep learning image classifier on benchmark image datasets. We begin with a standard baseline convolutional neural network (ResNet18) and then apply stronger augmentation techniques to see whether they improve generalization performance on clean test data and, later, on corrupted versions of the data.

Our long-term goal is to compare classical augmentations (random crop, horizontal flip) with mix-based methods such as Mixup (and later CutMix) and, in the final stages, GAN-based augmentation on CIFAR-10. CIFAR-100 and SVHN will be added in subsequent phases to study how augmentation strategies transfer to datasets with higher class complexity and different image domains.

Project Plan and Phases

Phase 1 – Baseline pipeline (CIFAR-10)

- Load the CIFAR-10 dataset and apply standard augmentations (random crop with padding, random horizontal flip, tensor conversion).
- Train a baseline ResNet18 classifier from scratch on CIFAR-10 using these standard augmentations.
- Evaluate the model on the clean CIFAR-10 test set and record test accuracy as the reference performance level.

Phase 2 – Stronger augmentation (Mixup on CIFAR-10)

- Implement Mixup at the mini-batch level by linearly combining input images and their corresponding labels using a Beta-distributed mixing coefficient.
- Train a second ResNet18 model on CIFAR-10 using the same training setup as the baseline, but with Mixup applied during training.
- Evaluate this Mixup model on the clean CIFAR-10 test set and compare its test accuracy to the baseline to see whether Mixup improves or degrades performance under the current settings.

Phase 3 – Robustness and additional datasets (planned)

- Extend the same baseline and Mixup training pipeline to CIFAR-100 and SVHN using consistent architectures and training protocols.
- Evaluate robustness using corrupted test data (e.g., CIFAR-10-C and other synthetic corruptions) to understand how different augmentations affect natural corruption robustness.
- Analyze which augmentation combinations best balance accuracy and robustness across datasets.

Phase 4 – GAN-based augmentation on CIFAR-10 (planned)

- Train a class-conditional GAN on CIFAR-10 to generate additional synthetic images.
- Integrate GAN-generated samples into the training set alongside classical and mix-based augmentations.
- Compare models trained with and without GAN-based augmentation in terms of both accuracy and robustness.

For this first midway submission, we focused on CIFAR-10 only in order to build and validate the full training pipeline end-to-end before scaling to other datasets and robustness experiments.

Phase 1 – Baseline Model Implementation (Completed)

In Phase 1, we implemented the baseline training pipeline on CIFAR-10 using ResNet18.

- Model: ResNet18 with the final fully connected layer modified to output 10 classes corresponding to CIFAR-10 categories.
- Data preprocessing and augmentation (train):
 - RandomCrop(32, padding=4)
 - RandomHorizontalFlip()
 - ToTensor()
- Data preprocessing (test):
 - ToTensor() only (no augmentation on the test set).
- Optimization setup:
 - Loss function: Cross-entropy loss
 - Optimizer: Adam with learning rate 0.001
 - Device: GPU (CUDA) when available.

We trained the baseline model for 5 epochs on CIFAR-10. After five training epochs, the baseline ResNet18 reached 69.8% test accuracy on the CIFAR-10 test set (as recorded in our report text), which provides a starting point for evaluating whether stronger augmentations like Mixup can further improve or hurt performance under the same training budget.

Phase 2 – Mixup Implementation (Completed for CIFAR-10)

In Phase 2, we introduced Mixup as a stronger data augmentation technique and trained a second ResNet18 model on CIFAR-10 with it. Instead of using each training image on its own, Mixup blends two images together and also blends their labels, so the model learns from “in-between” examples that lie between two classes. At a high level, for every mini-batch we randomly pick pairs of images, create a mixed image that visually looks like a combination of the two, and then train the model to give a prediction that is also a combination of the two corresponding labels.

We kept the model architecture and optimization setup the same as in Phase 1: a new ResNet18 model, trained from scratch on CIFAR-10, using cross-entropy loss and the Adam optimizer with a learning rate of 0.001. Training was run for 5 epochs, and Mixup was applied to every mini-batch during training so that the model consistently sees mixed images instead of only the original ones. After these 5 epochs, the Mixup model reached 55.9% test accuracy on the CIFAR-10 test set. With this short training schedule and basic hyperparameters, Mixup actually performed worse than the baseline, suggesting that we may need to adjust the training settings (for example, train longer or tune the Mixup strength) before it can show its potential benefits.

Comparison of Preliminary Results

We summarize the preliminary CIFAR-10 results obtained so far:

Model / Augmentation	Dataset	Epochs	Test Accuracy
Baseline ResNet18 with standard augmentation (RandomCrop + HorizontalFlip)	CIFAR-10	5	69.8%
ResNet18 with Mixup augmentation	CIFAR-10	5	55.9%

At this early stage, Mixup does not improve accuracy compared to the baseline; the reduced performance suggests that our Mixup configuration (mixing strength, number of

epochs, and possibly learning rate schedule) may not yet be optimal for this short training run. In later phases, we plan to:

- Increase training epochs and tune the Mixup hyperparameter alpha.
- Compare additional augmentations such as CutMix.
- Evaluate robustness on corrupted data to see whether Mixup offers benefits in robustness even if clean test accuracy is not higher.

Status Summary and Next Steps

Completed so far (First Midway):

- Implemented an end-to-end CIFAR-10 training pipeline with ResNet18 and standard augmentations.
- Trained the baseline model and obtained a clean test accuracy of 69.8% after 5 epochs.
- Implemented Mixup and trained a second ResNet18 model for 5 epochs, obtaining 55.9% test accuracy.
- Verified that data loading, augmentation, training, and evaluation all work correctly on GPU.

Planned for upcoming phases (towards Second Midway and final project):

- Extend the baseline and Mixup pipelines to CIFAR-100 and SVHN.
- Conduct robustness experiments using corrupted data (e.g., CIFAR-10-C) for both baseline and augmented models.
- Introduce additional strong augmentations (e.g., CutMix) and compare their impact on accuracy and robustness.
- Implement and train a class-conditional GAN on CIFAR-10 and integrate GAN-generated samples into the training process.